

Project 2

Operating Systems

Notice for Rust

I do not recommend that you use Rust for this project.

- Lack of time is not worthwhile for the effort and complexity.

Deadlines

Project 2 Group Signup and Division of Labor:

- Everyone can rebuild their group.
- You can register your groups in Canvas -> people -> Project 2
- Project Group Grading Policies: refer to General Project Syllabus.
- **Wednesday, October 8, 2025**

Project 2 Due Date:

- **Monday, November 3, 2025**
- Students that are not signed up before the deadline will be assigned to a group randomly.

Grouping

- Students that are not signed up before the deadline will be assigned to a group randomly.
- If you would like to work solely or working in a 2-student group, please email cop4610t@cs.fsu.edu to get the permit.
- If your group is a sole group or a 2-student group, please email me (cop4610t@cs.fsu.edu) in advance so that we will not randomly assign students to your group.
- **NOTE:** Please use your CS email to send an email to TA.

Group Changes

Please email me (cop4610t@cs.fsu.edu) if you would like to change the group after signup (Oct. 8th).

Group change requests are only accepted by Oct. 10th.

Join Project 2 GitHub Classroom

Click the link to the invitation page

Prof. Andy Wang Class Students -> <https://classroom.github.com/a/pXl8zO-U>

Prof. Xin Liu Class Students -> <https://classroom.github.com/a/3hSF7760>

Join Project 2 GitHub Classroom

The team leader needs to create a new team for Project 2. Please use the format of “project-2-group-xx” for your team's name. Scrolling all the way down to the bottom of this page to find the “Create team” button.

Team members join the created team accordingly.

Do **NOT** join the groups created for Project 1 and created by other teams.

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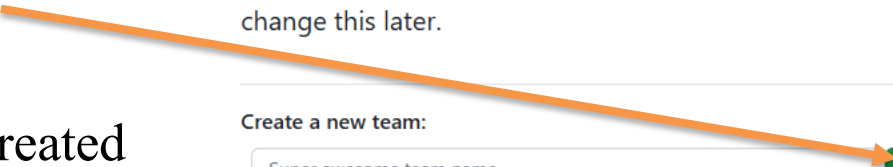
Accept the group assignment — Project 2

Before you can accept this assignment, you must create or join a team. Be sure to select the correct team as you won't be able to change this later.

Create a new team:

Super awesome team name

+ Create team



Overview

The project is split up into three parts:

Part 1: Tracing System Calls

- You will trace 4 system calls and use a diagnostic tool strace to generate debugging information of your system calls.

Part 2: Timer Kernel Module

- You will create a kernel module that writes the elapsed time to the proc file every call to the proc file and displays the current time of the system.

Part 3: Elevator Kernel Module

- You will create a kernel module that simulates an elevator. This elevator will abide to certain rules and writes to the proc file.

Project Breakdown

Part 1: Tracing System calls (~ 5 mins, can be done on Linprog)

Part 2: my_timer Kernel Module (~ half week, require sudo privilege)

Part 3: elevator Kernel Module (~2 weeks, require compiling kernel)

Note: it is important that you have compiled the kernel by Oct. 15th!
Otherwise, you will not have time to complete! Please attend office hours if you are struggling to compile the kernel with your system calls.
Otherwise, we may NOT be able to help at the last minute.

Development Environment

Virtual Machines	Lab Machines
+ Easier development (reboot VM vs. whole machine). + On your own computer. + Can save OS snapshots (virtual box only). - VM overhead (performance). - Must set up VM and install Ubuntu yourself. - Need a good computer for compiling kernel (≥ 4 cores). - Everyone has to install their own custom kernel.	+ Ubuntu already installed. + Group members can share same computer and same kernel. + Remote desktop access. - GUI interface may be slow. - May be slow depending on number of students using lab machines.

Requirements

- ✓ Computer with Root Access.
- ✓ Linux Kernel Version 6.x.x.
- ✓ Computer with lot of CPU cores.
- ✓ At least 8 gigabytes of RAM.
- ✓ ≥ 70 GB of HDD space.

Lab Machines

Each group will have a lab virtual machine. These machines will be remote, and you will have access to a GUI that allows you to interact with the Desktop of that machine. You will share this machine with your entire group, but they have to use their CS account to login to that machine.

If you do not have an CS account, please refer to System Group's web to create one

<https://system.cs.fsu.edu/newuser/cs-account-setup/>

If your group requires a lab VM. Please email the TA cop4610t@cs.fsu.edu to request one. Group leader should include all your group members names, FSUID's, CS usernames in a single email and it send to us. VM requests are only accepted until **Oct 10th**

NOTE: Please use your CS email to send an email to TA.

Lab Machines

You can also use a terminal to access the lab machine. This kind of connection is stable and easy to use.

1. Login to the GUI-based VM desktop and open a terminal. Then, type “ip addr” to get the address of this VM, its should be “192.168.xx.xx”
2. Open the terminal on your local PC/laptop/desktop, and ssh to the CS domain (shell.cs.fsu). Then, login to the VM by “ssh [your-cs-id@192.168.xx.xx](#)”

ref: <https://system.cs.fsu.edu/newuser/ssh-how-to/>

Virtual Machine

If you decide to do this on your own PC, you should do it in a virtual machine, so that you won't brick your computer.

I recommend VMWare because of performance and compatibility reasons. Although, you do not have access to snapshots on the free version.

You may consider using VirtualBox for snapshots, which may be vital for this sort of project.

Setup

We first need to pick a distro to use, for simplicity, we will use Ubuntu 24.04.1 LTS (same as the lab machines).

<https://ubuntu.com/download/desktop>

This distro comes with Linux Kernel version 6.8.x as the default (this is enough to run part 2).

We will be modifying the source of kernel version 6.10.x and installing it onto the same machine later on (for part 3).

Kernel Versions

NOTE: different versions of the kernel may have different APIs/codebase, which may not exactly as the same as project slides show.

Linux Kernel

- You will download the source of the latest stable release of the latest kernel, where you will be modifying by adding custom system calls.
- You can download the kernel from this website:
<https://www.kernel.org/>
- This is for part 3 of the project, where you will be constructing an elevator that transport pets to the correct floor.

Tips

- ❖ Start early!
- ❖ Have someone work on adding system calls and compiling the kernel while you work on the timer module.
- ❖ Attend office hours if you need any help. Here are office hours:

Majors Lab

Monday 2:00 – 3:00 PM (Bharadwaj)
Tuesday 3:00 – 5:00 PM (Rasheeq)
Wednesday 10:00 - 12:00 PM (Bharadwaj)
Thursday 3:00 – 4:00 PM (Rasheeq)
Friday 2:30 - 4:30 PM (Michael)